

OVERVIEW

Executive Summary

> **What we're building:** A single-page React app that lets children aged 5–8 with motor disabilities experience Kim Hoàng Vietnamese folk painting through a 7-screen, tap-only flow — choose paper color → stamp a woodblock outline → color zones → add a poem character → save the result as a PNG.

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The Core Bet

One tap = one meaningful creative action. No drag, no precision grip, no sustained pressure — just intention → result.

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Cultural Mission

Kim Hoàng painting is nearly extinct (last practised 1945, restored 2015). This app introduces a new generation to the authentic technique.

✅

MVP Deliverable

All 7 screens navigable. Pig motif fully functional. Rule-based AI color suggestions. Save as PNG. No backend required.

CULTURAL FOUNDATION

What Is Kim Hoàng Painting?

Tranh Kim Hoàng is an 18th-century Vietnamese woodcut folk art from Kim Hoàng village, Hoài Đức, Hanoi. Understanding the tradition is essential to understanding every design decision in this app.

1

Pre-dye the paper

Paper is dyed scarlet red (giấy hồng điều) or Chinese yellow (giấy vàng tầu) before anything else. The background carries emotional weight.

2

Stamp the woodblock

A carved jackfruit wood block is inked and pressed onto the paper — creating the bold black outline in one decisive action.

3

Hand-color zones

Mineral-based pigments (vermilion, gardenia yellow, white gypsum, indigo) are applied by hand with straw brushes. Same template, unique coloring every time.

4

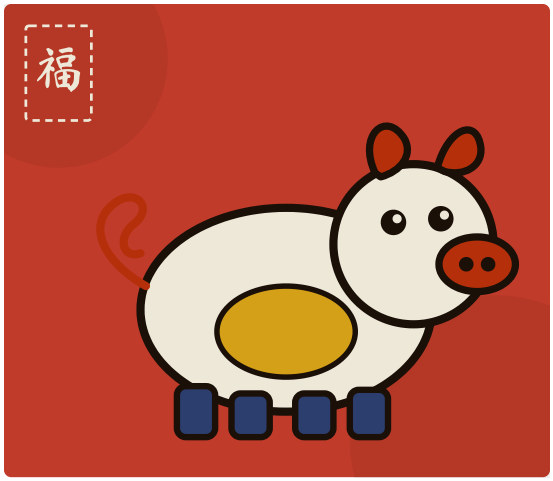
Write the poem

A Chinese character (福 Phúc, 祿 Lộc, 壽 Thọ) is handwritten in the top-left corner in cursive chữ thảo script — image + word = complete meaning.

HISTORY

Nearly Lost

A 1915 flood destroyed most original woodblocks in Kim Hoàng village. By 1945, no families continued the practice. Restoration efforts by collector Nguyễn Thị Thu Hoa began in 2015, recovering ~80 surviving paintings. The app participates in this cultural continuity.



App-generated Kim Hoàng pig (Lợn) on scarlet paper — matching authentic motif style. The white body, red ears, indigo legs, and yellow belly are the historically accurate Kim Hoàng color scheme.

TECHNIQUE

Why the Pig?

The pig (lợn) is the most iconic Kim Hoàng motif — a Tết symbol of prosperity. The fat, round pig with a curled tail is drawn in a characteristic folk style: bold black outline, chunky proportions, flat mineral colors. It is the MVP's primary motif.

↗ Wikipedia: Kim Hoàng Painting

↗ Kim Hoàng: Folk Art on the Verge of Extinction

↗ Vietnam Tourism: Kim Hoàng Village

↗ Google Arts & Culture: Revival of Kim Hoàng

↗ Vietnam Fine Arts Museum

↗ YouTube: Tranh Kim Hoàng demos

PROBLEM & OPPORTUNITY

Who Is Excluded and Why

Traditional art activities — including village visits to Kim Hoàng — assume a set of motor capabilities that many children simply don't have.

> **The exclusion barrier:** Holding a brush, staying inside lines, applying consistent pressure, and using bilateral hand coordination are all required by traditional painting. Children with cerebral palsy, developmental coordination disorder (DCD), or upper-limb differences are systematically excluded from the creative experience.

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Primary User

- Age 5–8
- Motor disability (tremors, limited fine-motor control, single-finger tap only)
- May have co-occurring attention differences
- Likely uses a tablet with a caregiver nearby

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Secondary User (Facilitator)

- Parent, teacher, or therapist
- Sets up the app before handing it to the child
- May toggle caregiver settings (sound, mode)
- Sits alongside but does not control the session

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Design Non-Goals

- No drag interactions
- No voice control (co-occurring speech impairments)
- No time pressure or countdowns
- No "wrong answer" states
- No login or accounts

> **Research backing:** Design constraints derive from WHO Global Report on Assistive Technology (2021), UNICEF "Designing for Children with Disabilities"

> guidelines, and Universal Design for Learning (UDL) Framework. See References for full citations.

VISUAL IDENTITY

Design System

Every visual decision is derived from Kim Hoàng's physical materials — the scarlet paper, woodblock ink, and five mineral pigments. There is no neutral background: the whole app lives on red paper.

Kim Hoàng Pigment Palette

Giấy Đỏ	Giấy Vàng	Mực / Ink	Son Đỏ	Vàng	Trắng Thạch Cao	Chàm / Indigo
#C13B2A	#C89A1A	#1A1008	#B5300A	#D4A017	#EDE8D8	#2C3E6E

Gỗ Mít / Wood
#8B6340

Typography

DISPLAY — PLAYFAIR DISPLAY 700

Tranh Kim Hoàng

DISPLAY ITALIC — PLAYFAIR DISPLAY 700I

Bắt đầu vẽ!

BODY — BE VIETNAM PRO 400/600 (VIETNAMESE OPTIMIZED)

Chọn giấy nền · **Choose paper** · Proper diacritics: ắ ộ ư ề

CALLIGRAPHY — MA SHAN ZHENG (POEM CORNER CHARACTERS)

福 祿 壽

> **Component principle:** Buttons have a `4px solid block shadow` (woodblock press aesthetic). Tap state collapses the shadow 3→1px to simulate pressing a physical stamp. All interactive elements ≥ 80×80px.

USER EXPERIENCE

App Flow

A strict 7-screen linear flow. No branching, no dead ends. Back navigation available on screens 2–6 only. Every path leads to a completed painting.

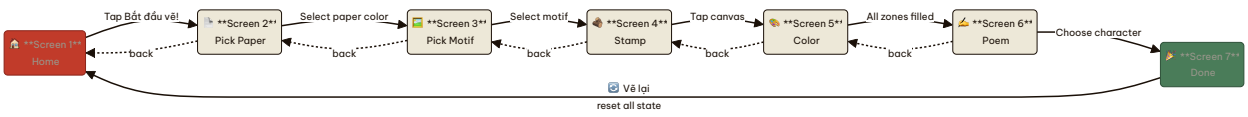


Fig 1 — Linear 7-screen UX flow. Dashed lines = back navigation. Green = terminal success state.

State Machine

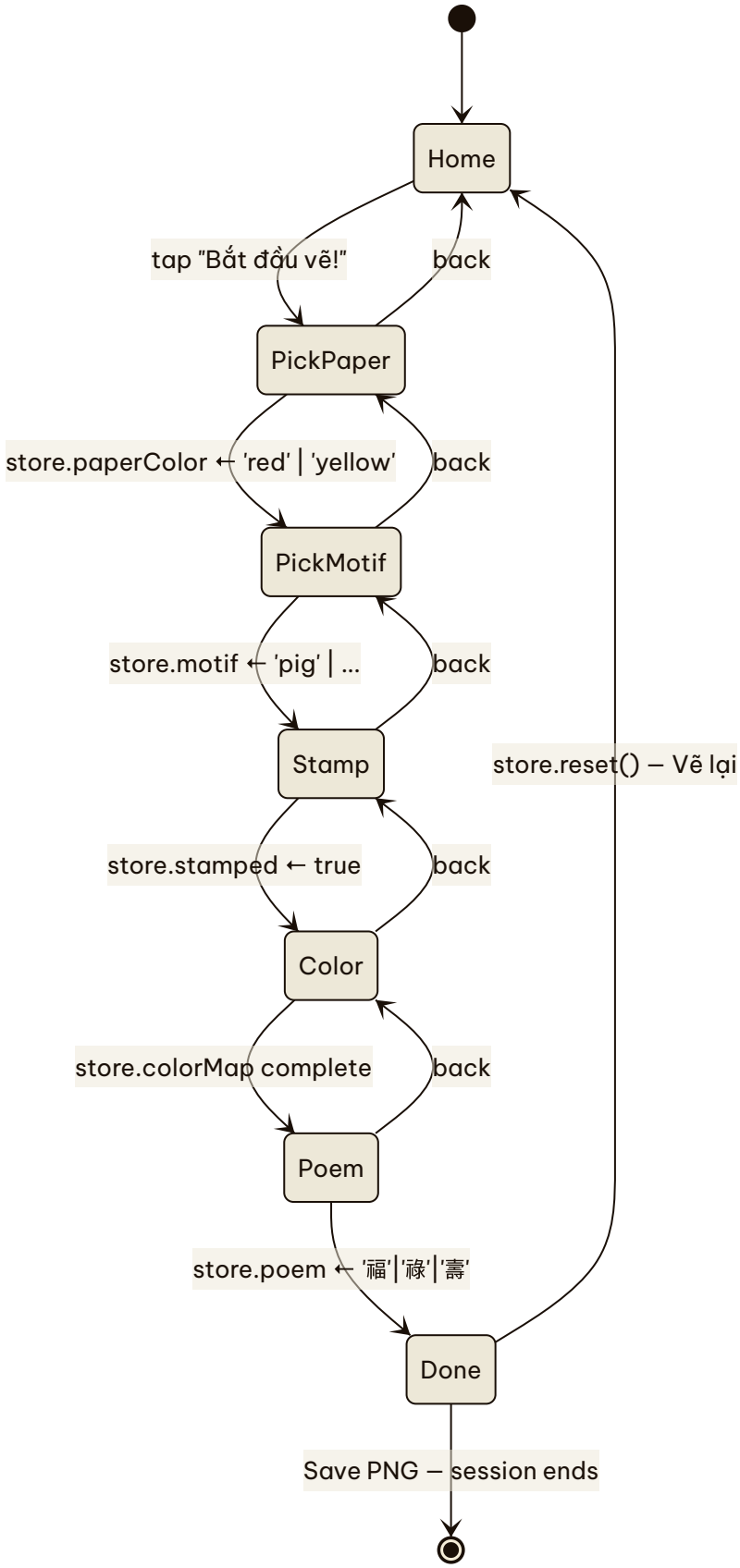

Home
tap "Bắt đầu vẽ!"
back
PickPaper
store.paperColor ← 'red' | 'yellow'
back
PickMotif
store.motif ← 'pig' | ...
back
Stamp
store.reset() — Vẽ lại
store.stamped ← true
back
Color
store.colorMap complete
back
Poem
store.poem ← '福'|'祿'|'壽'
Done
Save PNG — session ends

Fig 2 — App state machine mapping screens to Zustand store mutations.

SCREEN-BY-SCREEN

All 7 Screens

Each screen mockup shows the actual Kim Hoàng color palette and layout proportions at 240×420px (scaled phone view).

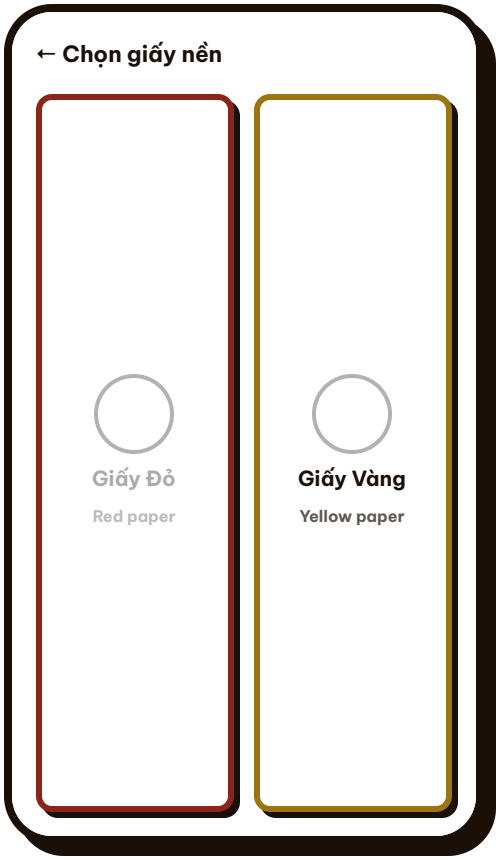
Screen 1 Home *Trang chủ*



Single entry point. The animated Kim Hoàng pig bounces gently on scarlet paper. One button, full width.

→ Tap button → slide-up transition to Screen 2

Screen 2 Pick Paper *Chọn giấy nền*



Two choices only. Authentic Kim Hoàng papers. Cards are ~160×200px — well above 80px minimum.

→ Tap card → spring bounce animation → navigate to Screen 3

→ No other tappable elements

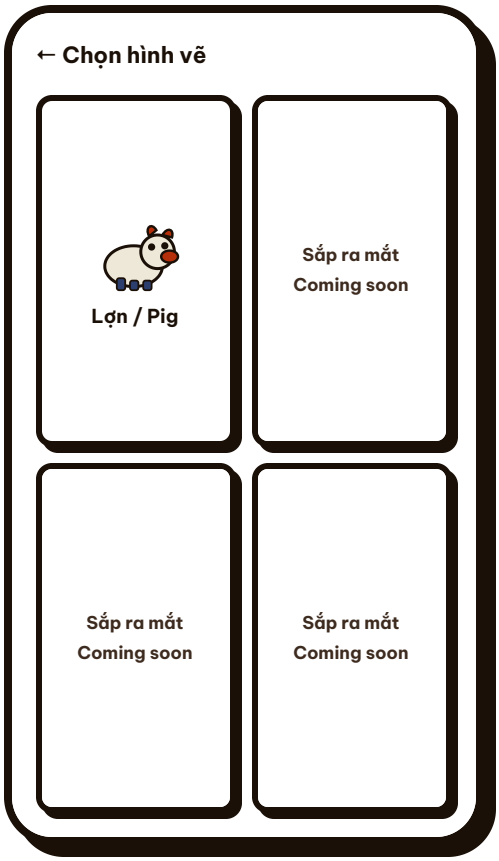
→ No state changes (stateless screen)

→ store.paperColor = 'red' | 'yellow'

→ Back arrow → Screen 1

`store.paperColor`

Screen 3 Pick Motif *Chọn hình vẽ*



4 choices, 2×2 grid. MVP: Pig is fully functional. Others show "Coming soon" overlay. Maximum cognitive load = 4 options.

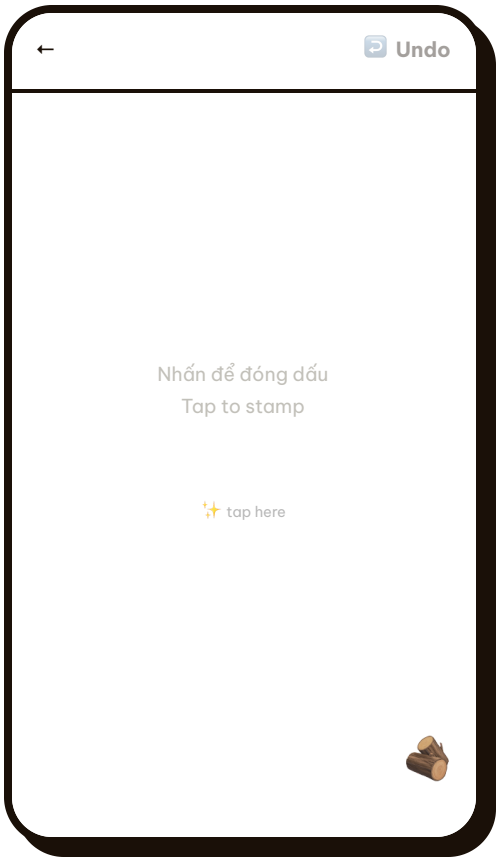
→ Tap pig card → bounce animation → Screen 4

→ store.motif = 'pig'

→ Other cards: tap shows overlay, no navigation

`store.motif`

Screen 4 The Stamp *Đóng dấu*



The culturally significant moment. The entire canvas is the tap target — maximum accessibility. Stamp presses down, then the pig outline draws itself with an ink-bleed animation.

→ Tap anywhere → stamp presses (scale + shadow, 300ms)

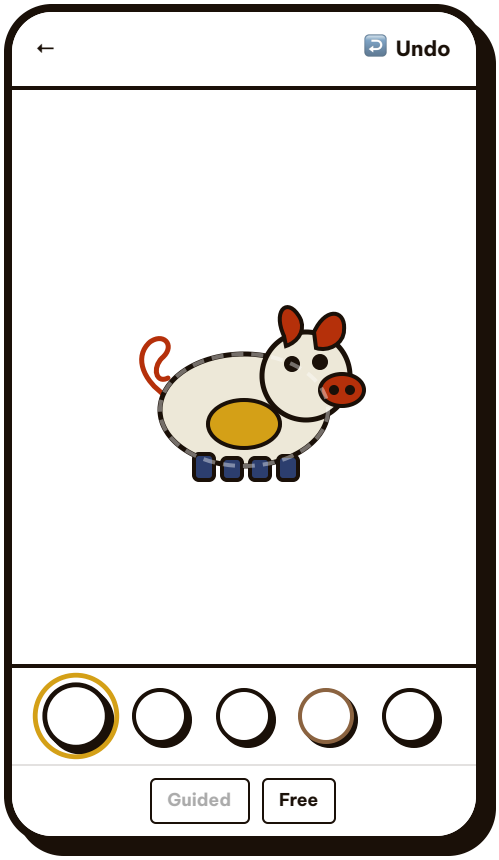
→ Pig SVG outline draws via pathLength 0→1 (600ms)

→ store.stamped = true

→ Auto-navigate to Screen 5 after 1200ms

`store.stamped = true`

Screen 5 Coloring *Tô màu*



Core experience. In Guided Mode the app suggests historically accurate colors. Zone snapping forgives imprecise taps within 80px. Undo always visible.

- → Tap zone → zone snapping finds nearest zone center (80px radius)
- → Guided: AI suggests Kim Hoàng color → floods in
- → Free: child picks from 5-color palette first
- → Miss → gentle canvas shake, no error text
- → All zones filled → "Xong / Done" button appears
- → Undo → reverts last fill from stack

```
store.colorMap { zone_id: hex }
```

Screen 6 Poem Corner *Góc thơ*



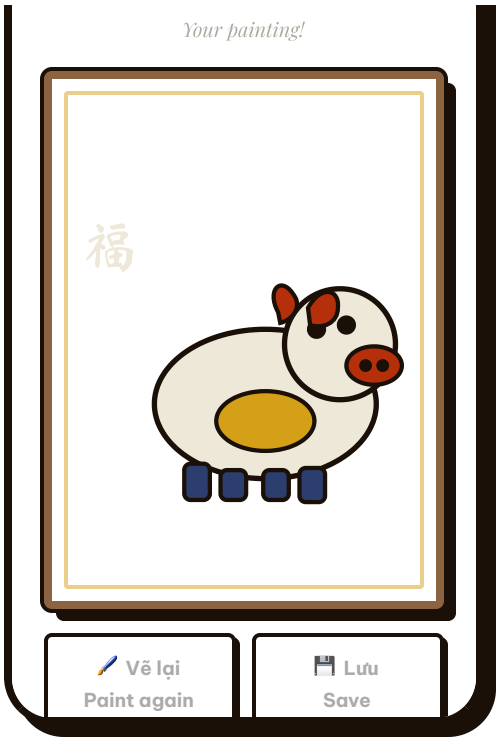
Closing ritual. The dotted box in the painting's top-left pulses to invite interaction. Tapping a character animates it into place with a brush-stroke calligraphy animation.

- → Dotted corner box pulses → visual affordance
- → Tap character → calligraphy draw animation into corner
- → Tapping another replaces it (no undo needed)
- → "Xong! Done ✓" appears after selection
- → store.poem = '福' | '祿' | '壽'

```
store.poem
```

Screen 7 Done *Hoàn thành!*





Celebration state. Confetti animation on mount. Painting displayed in a framed, "pinned-to-wall" style. Two clear actions.

- → Confetti animation on load (framer-motion)
- → "Vẽ lại" → store.reset() → navigate to Screen 1
- → "Lưu 💾" → export canvas as PNG (html2canvas or Web Share API)
- → Optional audio: narrates what the child painted

```
store.reset() | PNG export
```

INTELLIGENCE

AI Assist System — Three Layers

The "AI" in this app is intentionally rule-based for MVP — no live model calls, no API dependency. This is a deliberate accessibility design choice: intelligent defaults, not a language model.

1 Zone Snap Tolerance ✅ MVP

Children with tremors or imprecision aim for a zone and may land 50px off-center. The system finds the nearest zone center within an 80px snap radius and credits the tap to that zone. If no zone is within range, a gentle canvas shake plays — no error text, no red X. The snap radius can be increased to 120px for higher accessibility needs.

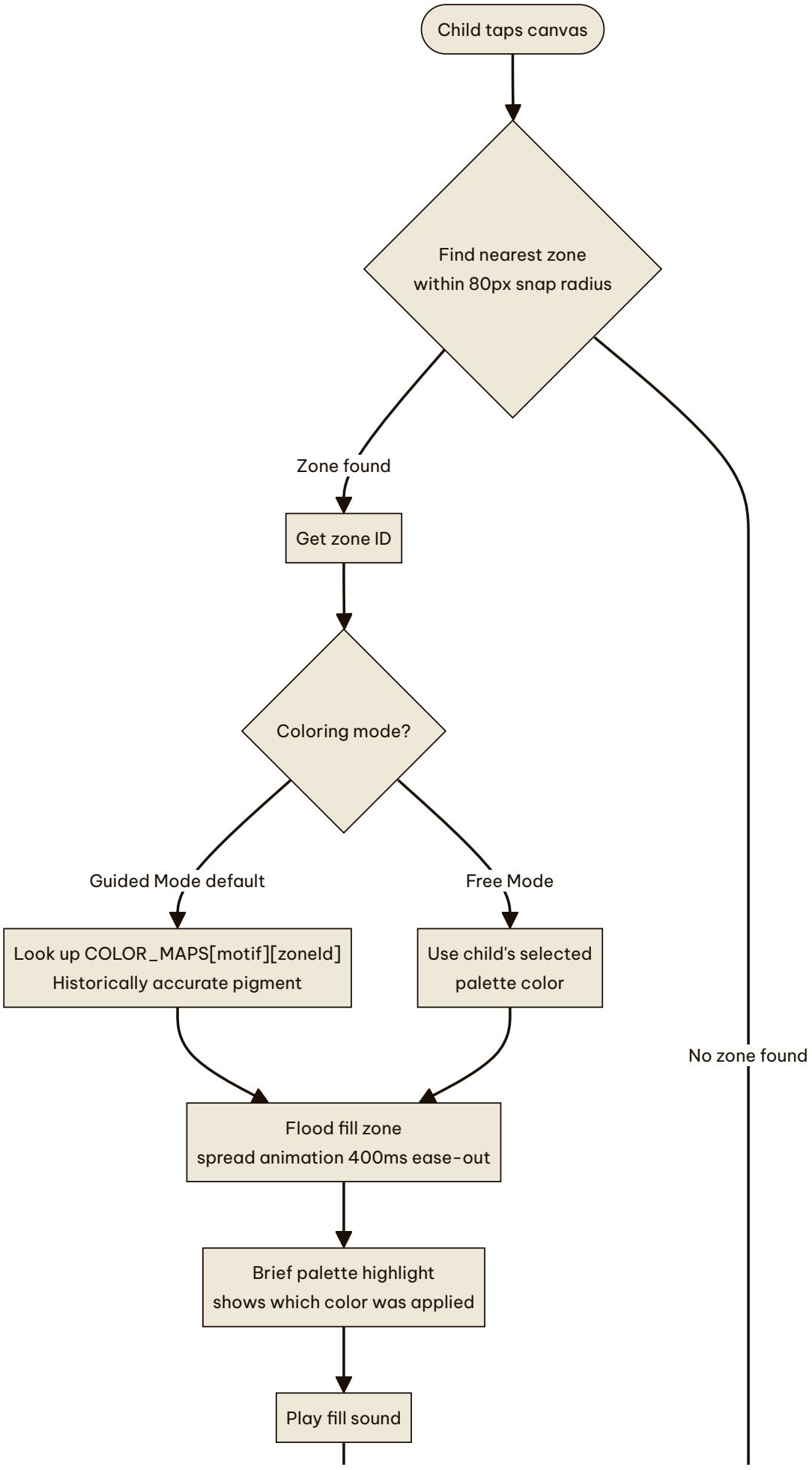
2 **Color Suggestion Engine** ✅ MVP

In Guided Mode, the app automatically selects the historically accurate Kim Hoàng color for each zone — derived from research on real 18th-century paintings. The child taps a zone and the correct color floods in without needing to make a palette choice. In Free Mode, the child manually selects a color first. Completion is detected when every zone in the motif's color map has been filled.

3 **Stroke Completion** ❌ V2 Only

Higher-functioning children submit a rough scribble; a fine-tuned vision model (ControlNet scribble mode or Claude API with vision) returns a Kim Hoàng-style SVG outline. Out of scope for MVP due to API latency, cost, and portfolio demo constraints. Documented as the full product vision.

Coloring Interaction Flow


Child taps canvas
Find nearest zone
within 80px snap radius
Zone found
No zone found
Get zone ID
Coloring mode?
Guided Mode default
Free Mode
Look up COLOR_MAPS[motif][zoneId]
Historically accurate pigment
Use child's selected
palette color
Flood fill zone
spread animation 400ms ease-out
Brief palette highlight
shows which color was applied
Play fill sound

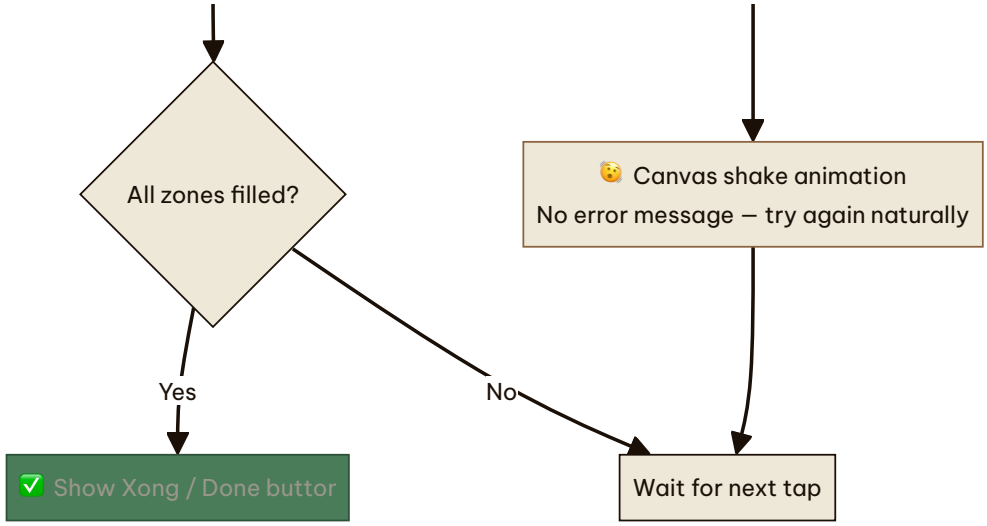


Fig 3 — Complete coloring interaction decision tree including zone snap and mode branching.

Traditional Color Map (Pig — Lợn)

Zone ID	Zone Name	Color	Pigment (Vietnamese)	Hex
body	Body	White gypsum	Trắng thạch cao	#EDE8D8
ear-left, ear-right	Ears	Vermilion	Son đỏ	#B5300A
eye-left, eye-right	Eyes	Chinese ink	Mực tàu	#1A1008
snout	Snout	Vermilion	Son đỏ	#B5300A
tail	Tail	Vermilion	Son đỏ	#B5300A
leg-1 through leg-4	Legs	Indigo	Chàm	#2C3E6E
belly-spot	Belly	Gardenia yellow	Vàng hoa dành dành	#D4A017

INCLUSIVE DESIGN

Accessibility Commitments

Every constraint in this app is an accessibility decision. These are non-negotiable requirements, not polish.

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Tap Only — No Drag

Zero drag interactions anywhere in the app. Children with tremors or limited grip cannot reliably drag. One physical gesture = one creative action.

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80px Minimum Tap Targets

Every tappable element is at least 80×80px. Most are significantly larger (the stamp canvas is the entire screen). Enforced in every component.

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No Time Pressure

No countdowns, no timeouts, no loading bars that run out. The child can take as long as they need on every screen.

✅

Every Tap Produces a Result

Nothing silently fails. A tap on a zone produces a fill. A miss produces a shake. The stamp tap produces an animation. Failure is always soft and visual.

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Color + Shape, Never Color Alone

No information is conveyed by color only. Every color swatch has a Vietnamese label. Every zone has a shape. Text contrast ≥ 4.5:1 on all backgrounds.

🇻🇳

Bilingual (Vietnamese + English)

Every user-facing string has Vietnamese primary label and English secondary. Uses Be Vietnam Pro — the only major sans-serif optimized for Vietnamese diacritics.

↩️

Undo Always Visible

One large undo button, always visible during coloring. Maintains a stack of zone fills. Children can explore without fear of ruining their work.

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Reduced Motion Support

All animations respect `prefers-reduced-motion`. When enabled, transitions collapse to instant state changes. No flashing or strobing effects anywhere.

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Success Is Guaranteed

Guided Mode's color suggestions ensure every child produces an authentically Kim Hoàng

painting — regardless of color knowledge or decision-making ability.

TECHNICAL

Architecture & Tech Stack

Client-side only. No backend, no API calls, no authentication. Runs in any modern browser.

`React 18` `TypeScript` `Vite` `react-router-dom v6` `zustand` `framer-motion` `konva + react-konva` `html2canvas` `No backend` `No API calls` `No app store`

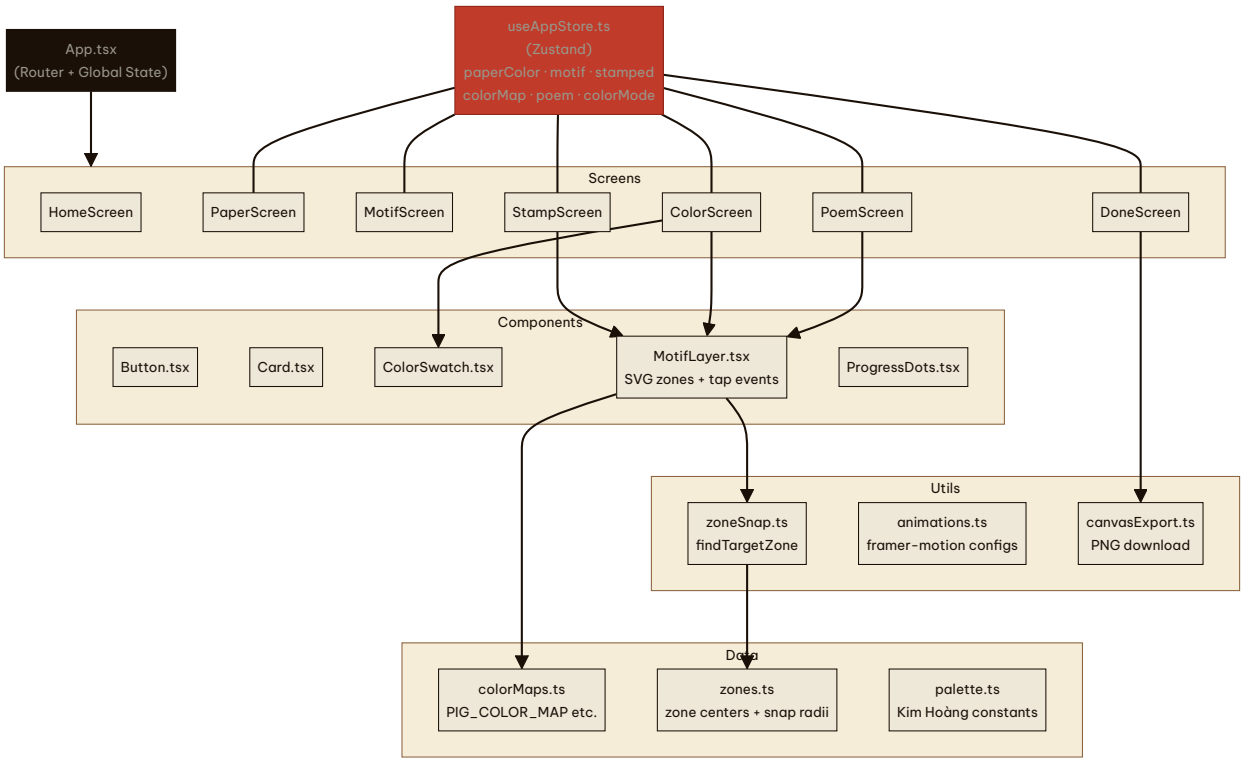


Fig 4 — Component and data architecture. Zustand store is the single source of truth shared across all screens.

> **No TypeScript `any`.** The codebase enforces strict TypeScript throughout. All zone IDs, motif names, paper colors, and poem characters are typed as union literals — no magic strings.

BUILD PLAN

MVP Scope & Phasing

9 phases in dependency order. Each phase is independently demoable before moving on.

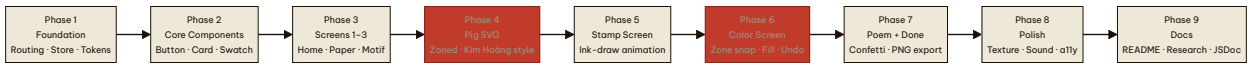


Fig 5 — Build phases in dependency order. Red = highest-risk phases (SVG asset + core coloring UX).

Phase 1	**Foundation** CSS tokens, Zustand store, react-router-dom routing, all 7 screen stubs. Verification: all routes load, no errors.	Core
Phase 2	**Core Components** Button, Card, ColorSwatch, ProgressDots — all with accessibility props and framer-motion animations.	Core
Phase 3	**Screens 1–3**	Core

	Home, Paper selection, Motif selection — fully functional. First demoable milestone.	
Phase 4	**Pig SVG (Critical Asset)** Kim Hoàng-style pig SVG with 12 individually-ID'd zones matching PIG_COLOR_MAP. Zone center coordinates measured for snap logic.	Core · High Risk
Phase 5	**Stamp Screen** Full tap-to-stamp with SVG pathLength ink-draw animation (50ms stagger per path). Auto-advance after 1200ms.	Feature
Phase 6	**Color Screen** Zone snap, Guided + Free modes, fill animation, undo stack, completion detection. Core UX of the app.	Feature · High Risk
Phase 7	**Poem + Done** Calligraphy animation, confetti, PNG export via html2canvas. Full end-to-end flow complete.	Feature
Phase 8	**Polish** Paper grain texture, optional sound effects with caregiver toggle, prefers-reduced-motion, responsive testing (375px / 768px / 1024px).	Polish
Phase 9	**Documentation** RESEARCH.md, README.md, component JSDoc comments explaining every accessibility decision for portfolio evaluators.	Docs

> **MVP Definition of Done:** All 7 screens navigable · Pig motif stamp → color → poem → save working end-to-end · Zone snapping functional · Save produces a real PNG file · Loads on mobile Chrome/Safari without errors.

SOURCES

References & Further Reading

CULTURAL · PRIMARY REFERENCE

Wikipedia — Kim Hoàng Painting — History, technique, motifs, near-extinction, and 2015 restoration. Starting point for all cultural research.

CULTURAL · MUSEUM

Vietnam Fine Arts Museum (Bảo tàng Mỹ thuật Việt Nam) — Holds original Kim Hoàng works. Authentic reference for motif style, proportions, and color palette.

CULTURAL · ARTICLE

Google Arts & Culture — Revival of Kim Hoàng Folk Paintings — Documents Nguyễn Thị Thu Hoa's restoration work and surviving painting catalog.

CULTURAL · TOURISM

Vietnam National Administration of Tourism — Kim Hoàng Village — Village location, artisan context, and cultural significance for Tết.

ACCESSIBILITY · GLOBAL REPORT

WHO — Global Report on Assistive Technology (2022) — Motor disability prevalence statistics and assistive design principles for children.

ACCESSIBILITY · FRAMEWORK

CAST — Universal Design for Learning (UDL) Framework — Principles for multiple means of action and expression. Grounds the tap-only interaction model.

ACCESSIBILITY · GUIDELINES

UNICEF — Designing for Children with Disabilities — Specific design guidelines for motor-impaired children aged 5–10.

TECHNICAL · ANIMATION

Framer Motion Documentation — Spring physics, pathLength, whileTap. Used for all interactive animations in the app.

TECHNICAL · CANVAS

React Konva Documentation — Canvas-based zone rendering and tap event handling.

TECHNICAL · EXPORT

html2canvas — DOM-to-canvas rendering used for PNG export of the final painting.